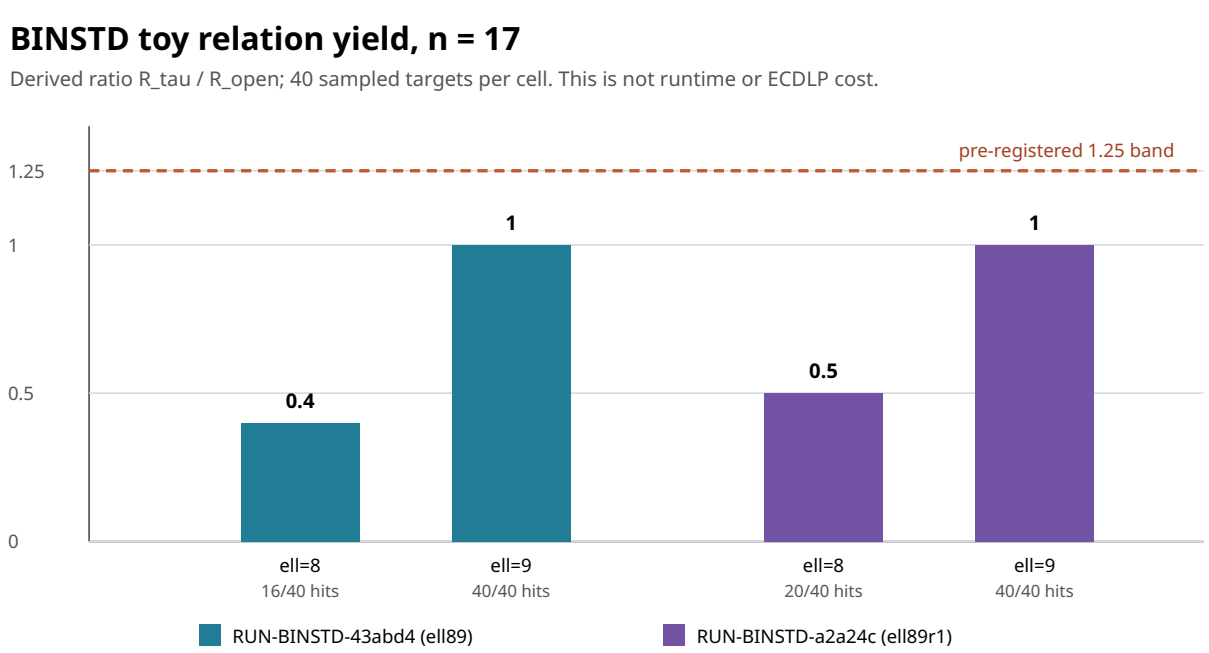
Source-pinned replay of BINSTD ell89 toy panels

2026-10-22 EXP-BINSTD-8bdf86 H-BINSTD-5a212b

This validation replay used the historical code commits recorded in the two run environments and copied their hash-matched Stage-0 freeze files into isolated temporary directories. It did not write to any frozen run record. This is not a new experiment or evidence decision.

Panel / Stage-1 run	ℓ	Tau hits	Open hits	R_τ/R_{open}	Band
ell89 / RUN-BINSTD-43abd4	8	16/40	40/40	0.4	1.25
ell89 / RUN-BINSTD-43abd4	9	40/40	40/40	1.0	1.25
ell89r1 / RUN-BINSTD-a2a24c	8	20/40	40/40	0.5	1.25
ell89r1 / RUN-BINSTD-a2a24c	9	40/40	40/40	1.0	1.25



Both freshly generated yield matrices, admissible-dimension files, and results notes match the archived bytes exactly. Raw-result payloads match after excluding the new timestamps. The source files, dated specifications, and Stage-0 freeze artifacts match their recorded SHA-256 pins. The committed `receipt.json` lists all hashes; a second end-to-end invocation of the committed replay command reproduced it byte for byte. The first local replays took 355.8 and 315.6 seconds on Python 3.9.25; these are not benchmark timings. A one-count mutation in a copied matrix caused the verifier to fail before writing a receipt, including under `python3 -O`.

Scope. The four recorded toy ratios lie below the pre-registered 1.25 band at $n = 17$, $\ell \in \{8, 9\}$, under the exhaustive $m = 3$ pin. The open arm hit all 40 targets in each cell, so its observed rate was 1; the tau rate cannot exceed 1, making the 1.25 ratio unattainable on those measured cells. This is a ceiling limit on the tested protocol. An independent polynomial irreducibility calculation passed, and the degree-1 and two degree-8 factors of $x^{17} - 1$ multiplied out, giving subset sums $\{0, 1, 8, 9, 16, 17\}$. The normal-basis link to invariant subspaces has no Lean proof here. The pair-membership shortcut agreed with direct ordered triples on bounded samples. This is a same-source deterministic replay, not an independent elliptic-curve arithmetic oracle or Lean proof of the experiment. No ECDLP break, asymptotic transfer, runtime advantage, or rho comparison follows. No uncertainty interval was recorded for the 40-target hit rates.

Reproduce. Run `python3 docs/reports/2026-10-22-binstd-ell89-replay/replay.py -run` from the repository root. The detailed report and graph source are beside this PDF. The canonical performance graphs were checked; the replay creates no new speedup or attack-cost row.